

Understanding Network Performance of MicroVMs in Edge Systems for Autonomous Driving

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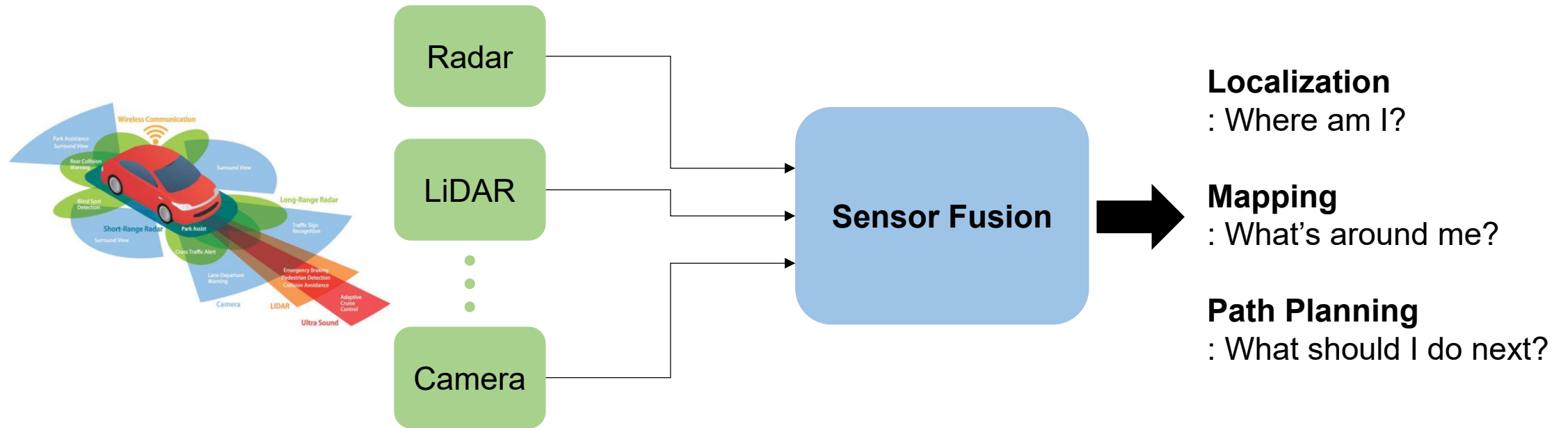
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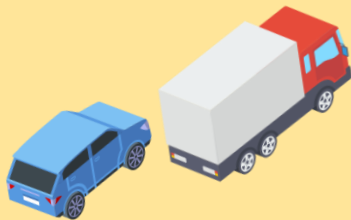
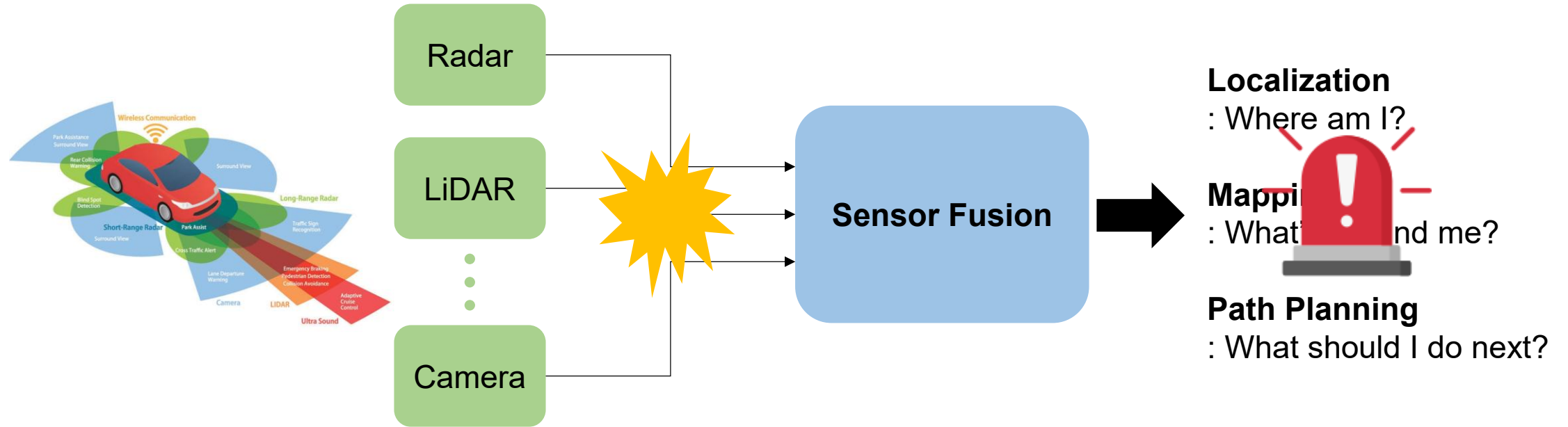
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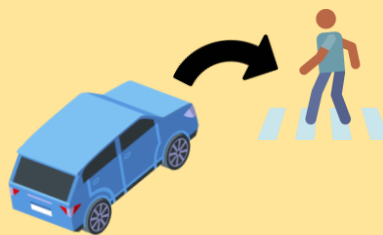
Standalone Autonomous Driving: **Architecture**



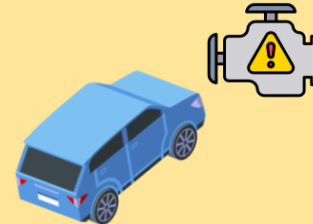
Standalone Autonomous Driving: **Limitations**



Occlusion



Blind spot



Sensor malfunction



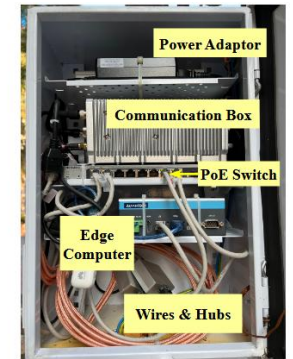
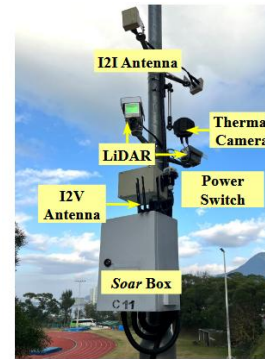
Severe weather

Cooperative and Edge-Assisted Autonomous Driving

- Safe autonomous driving cannot be fully achieved using only sensor data from individual vehicles.



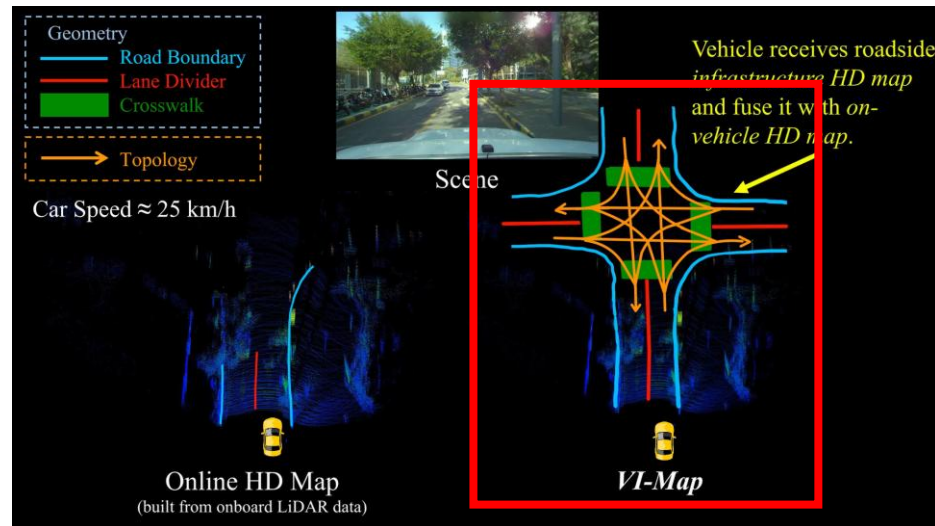
- **Connected Vehicles + Smart Roadside Infrastructure (SRI)**
 - Vehicles and smart edge systems (roadside units + edge computers) exchange real-time information such as traffic conditions, perception results.



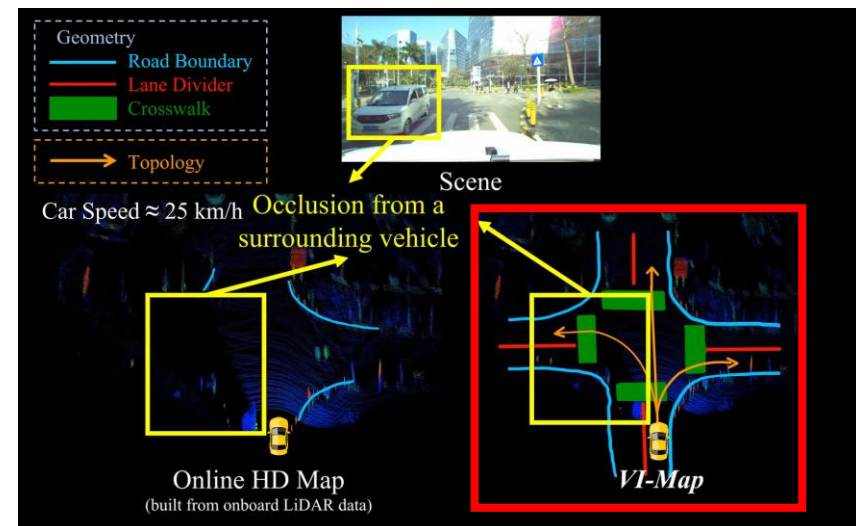
- Large-scale field trials have been conducted.
 - U.S. DOT CV Pilot, Shanghai's 5G-V2X, Europe's C-Roads, Korea's C-ITS
 - → Improved safety, traffic efficiency and reduced social costs

Representative Scenario: Perception sharing

- Infrastructure **share raw sensor data or processed results** to enable vehicles to construct a comprehensive scene and enhance their understanding of complex environments.



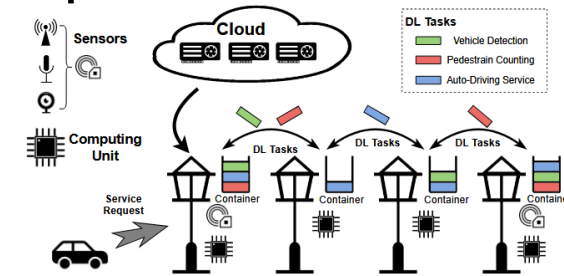
Better decision-making and routing



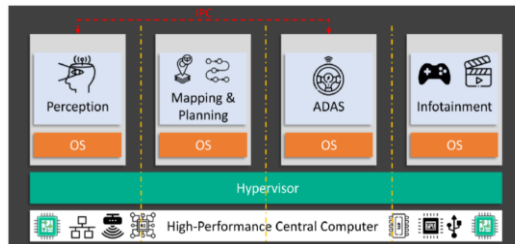
Safer driving without loss of information

Problem: Containers Are Fast, but Provide Weak Isolation

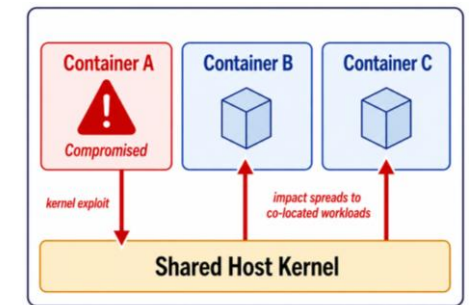
- SRI edge systems also use **containers** to deploy multiple workloads.
 - More capable edge devices + more complex workloads



- Automotive industry is transitioning toward SDVs (Software-Defined Vehicle).
 - Vehicle functions are decoupled from hardware and deployed/updated as software.
 - **Containers** are increasingly used as a runtime for SDVs.



- Critical weakness of containers: **shared host kernel**
 - A single container breach can affect all workloads on the same node.
 - This is particularly problematic in safety-critical scenarios.



Can MicroVMs Be an Alternative?

- Key requirements: **Network Performance** and **Isolation**
 - Low latency, high throughput
 - Predictable performance
 - Strong isolation under contention
- **MicroVMs** (e.g., Kata Containers, AWS Firecracker)
 - Container-like efficiency + VM-level isolation
- Challenges
 - Prior studies lack comprehensive analysis using realistic autonomous-driving workloads.
 - Extra processing layers may introduce latency and overhead.
 - MicroVM architectures vary depending on design goals (performance vs. minimal attack surface).



Research Question

“Can microVMs provide strong isolation for safety-critical workloads while meeting the network performance requirements of autonomous-driving edge systems?”

Benchmarks Based on Network Communication Patterns

I2I Communication

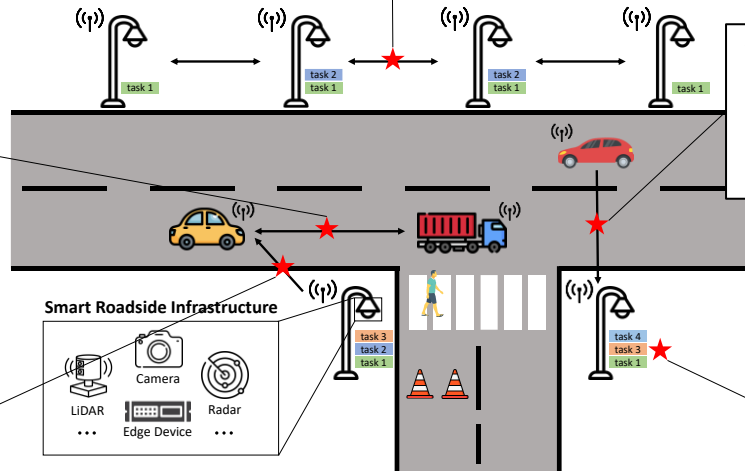
LiDAR point-cloud sharing between SRIs for sensor fusion
2MB frames @ 10Hz → Latency

V2V Communication

Safety message distribution via a broker
1KB-1MB messages @ 10Hz → Latency

V2I Communication

Vehicles query the edge and get a response.
1KB-1MB messages → Throughput, Latency



I2V Communication

Periodic data streaming to vehicles
(e.g., traffic conditions, perception outputs)
1KB-1MB messages @ 10, 100Hz → Throughput

Multi-tenant Contention

Interferer modes: CPU-saturating, network TX/RX
→ Performance degradation

Key Findings

- **Comparable throughput to containers, but at higher CPU cost.**
 - I2V: microVMs match container throughput across all message sizes, but consume up to **5.9× more system-level CPU time**.
 - Careful deployment is needed in resource-constrained edge systems.
- **MicroVMs can outperform containers in some scenarios.**
 - V2I: microVMs **outperform containers by up to 26%** for large payloads.
 - Batching across guest-host crossings improves efficiency.
- **Even between microVMs, architectural differences drive performance gaps.**
 - I2I: Firecracker's bursty delivery causes a **43.6% drop rate** (↔ Kata: 3.8%).
 - Multi-tenant contention: Firecracker degrades by **up to 77%** under network interference; Kata remains near baseline.

Conclusion

- **With workload-aware platform selection**, microVMs can be a viable alternative to containers in autonomous-driving edge systems.
- Performance depends on the interaction among **packet-processing architecture** × **traffic pattern** × **resource contention**.
- **Not all microVMs are equal**: design goals and implementation choices can lead to very different behavior under the same workload.
- We will optimize microVM network performance based on our analysis.